

Roll Number:.....

Thapar Institute of Engineering and Technology, Patiala
Department of Electronics and Communication Engineering

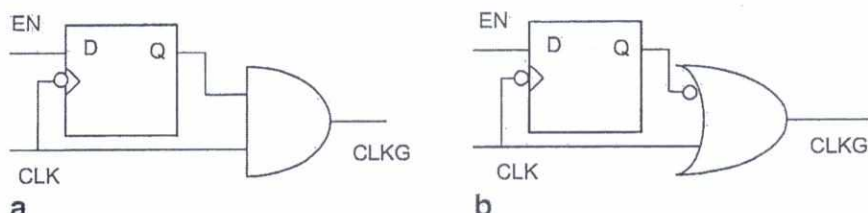
M.Tech.: EST- May 2025

Time: 3 Hours; MM: 45

PVL207: Low Power System Design

Name of Faculty: Dr. Rajneesh Sharma

Note: 1. Attempt any five out of six questions.
2. Take suitable assumption(s) if necessary.

Q.N.	Questions	Marks	CO	BL
Q.1	(a) Explain the working of a 6T SRAM cell in detail, covering all three fundamental operations: Read, Write, and Hold. (b) How the reduction of voltage swing on bit lines in SRAMs can be achieved to decrease the power consumption? Give detail explanation.	[5+4=9]	CO4	L1
Q.2	How does one-hot encoding contribute to power dissipation reduction in digital circuits? Analyze the underlying mechanisms and evaluate its effectiveness compared to other encoding techniques.	[5+4=9]	CO1	L3
Q.3	How does technology scaling influence various sources of power consumption in integrated circuits? Critically analyze the emerging low-power design methodologies developed to address power challenges specifically introduced by scaling trends.	[5+4=9]	CO3	L4
Q.4	Design a Moore state machine to detect the overlapping sequence '0110'. Discuss strategies for minimizing power consumption in your design. Additionally, provide a logic diagram that incorporates these power-saving techniques.	[5+4=9]	CO2	L4
Q.5	In what ways does the precomputation technique aid in reducing power consumption? Describe the implementation process and support your explanation with a suitable diagram.	[5+4=9]	CO1	L5
Q.6	Two level sensitive clock gating logics have been shown in Fig 1. Explain the working of both the logics and figure out which one is better for glitch suppression. Use the timing diagram for the explanation. 	[5+4=9]	CO2	L2